



Analgesic Effectiveness of Truncal Plane Blocks in Patients Undergoing the Nuss Procedure: A Randomized Controlled Trial

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ABSTRACT

Introduction: Most patients undergoing the Nuss procedure reported moderate to severe pain after surgery. This study aimed to investigate the efficacy and safety of ultrasound-guided serratus anterior plane block (SAPB) combined with transversus thoracic muscle plane (TTMP) block for relieving acute pain in patients undergoing the Nuss procedure.

Methods: The enrolled patients in our study were allocated to either receive combined nerve blocks with ropivacaine (NB group) or saline (CON group). The primary outcome of this study was postoperative pain at 2, 4, 8, 16, 24, 36, and 48 h during rest and movement (coughing). Secondary outcomes included intraoperative dosage

of remifentanyl, the time to extubation and the length of stay in the post-anesthesia care unit (PACU), the total acetaminophen and codeine tablet consumption, time to first bowel movement, time to first flatus, opioid-related adverse events, and the length of hospital stay.

Results: Patients in the NB group had significantly lower Numerical Rating Scale (NRS) pain scores compared with the CON group. The NB group required significantly less postoperative acetaminophen consumption and lower dosages of perioperative sufentanyl and remifentanyl compared with the CON group. The length of stay in the PACU and time to extubation were significantly increased in the CON group compared with the NB group. Time to first bowel movement and time to first flatus were earlier in the NB group. But there were no significant differences between the groups in terms of the length of hospital stay and codeine tablet consumption.

Conclusion: Ultrasound-guided SAPB and TTMP blocks in patients undergoing the Nuss procedure could provide effective analgesia.

Trial Registration: This study was registered in the Chinese Clinical Trial Registry (ChiCTR2000038506).

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Keywords: The Nuss procedure; Transversus thoracic muscle plane block; Sufentanyl; Serratus anterior plane block; Postoperative pain

Key Summary Points

Why carry out this study?

Although the Nuss procedure is minimally invasive with less tissue damage, the forced placement of a bar behind the anterior chest wall to elevate the sternal depression and correct the deformity of the chest wall is associated with significant postoperative pain during the first 2 postoperative days. Most patients undergoing the Nuss procedure report moderate to severe pain after surgery, which can limit mobility, reduce patient satisfaction, and cause longer hospital stays. For these reasons, we are looking for the best form of perioperative analgesia for patients with pectus excavatum.

What was learned from the study?

The present study showed that bilateral serratus anterior plane block (SAPB) combined with transversus thoracic muscle plane (TTMP) block could provide effective analgesia for 24 h after surgery, reduce perioperative sufentanyl and remifentanyl consumption, promote early recovery of intestinal function, and decrease the time to extubation and the length of stay in the post-anesthesia care unit (PACU) in patients undergoing the Nuss procedure.

the pectoralis major and rectus abdominis muscles, increased length of hospital stay, and complications [3]. Although the Nuss procedure is minimally invasive with less tissue damage, the forced placement of a bar behind the anterior chest wall to elevate the sternal depression and correct the deformity of the chest wall is associated with significant postoperative pain during the first 2 postoperative days [4, 5].

Most patients undergoing the Nuss procedure report moderate to severe pain after surgery, which can limit mobility, reduce patient satisfaction, and cause longer hospital stays [6]. For these reasons, anesthesiologists are looking for the best form of perioperative analgesia for patients with pectus excavatum. Thoracic epidural anesthesia was the most commonly used postoperative analgesia method in this population in the past [7]. But epidural hematoma, abscess, hemodynamic instability, and peripheral nerve lesions have restricted its application [8]. Video-assisted intercostal nerve cryoablation [9], continuous wound catheters delivering local anesthetic [10], bilateral thoracic paravertebral block [11], the erector spinae plane block [12], serratus anterior plane block (SAPB) [13], and pectoral nerve block [14] showed improvement in pain scores recently. Intercostal nerve cryoablation has the risk of the experiencing of neuropathic pain and prolonged numbness [15], and continuous wound catheters delivering local anesthetic pose a risk of catheter blockage [10]. There is a risk of hematoma and pneumothorax due to bilateral thoracic paravertebral block and the erector spinae plane block [16]. SAPB and pectoral nerve block are superficial and safe, but they cannot cover anterior branches of intercostal nerves to relieve pain at the site of the internal mammary area.

The transversus thoracic muscle plane (TTMP) block covers the internal mammary area and blocks the anterior branches of intercostal nerves [17], and SAPB covers the lateral cortex of the intercostal nerve from T2 to T9. Therefore, the combination of SAPB and TTMP block may provide effective analgesia for the Nuss procedure. The aim of our study was to assess whether SAPB combined with TTMP block can provide effective analgesia for patients undergoing the Nuss procedure.

INTRODUCTION

Pectus excavatum is the most common chest wall malformation, and the incidence rate is approximately 1 in 400 in young adults and children [1]. The minimally invasive repair of pectus excavatum (the Nuss procedure) is now considered to be the most common operation used to repair pectus excavatum [2] compared with the traditional open surgical technique (Ravitch), which can cause more intraoperative blood loss, longer surgical time, more damage to

METHODS

The ethics committee of the First Affiliated Hospital of Nanchang University approved our trial. Our study was registered at the Chinese clinical trial registry (ChiCTR2000038506), and written informed consent was obtained from the parents. The trial was conducted in accordance with the Declaration of Helsinki.

Patients and Design

This randomized, double-blind study was performed on patients with a diagnosis of pectus excavatum, and all participants underwent Nuss procedures from 1 February 2021 to 30 January 2023. Patients aged between 8 and 18 years with American Society of Anesthesiologists Physical Status I to II were included in our trial.

Exclusion criteria included previous thoracic surgery, history of neuropathic pain, allergies to study drugs, infection at the puncture site, inability to consent to the study, and long-term opioid use. Our study randomly divided all participants undergoing the Nuss procedure into two groups: the NB group, receiving bilateral TTMP block combined with SAPB, and CON group, receiving the same block with saline.

Randomization and Blinding

Our study used a computer program to perform a random allocation sequence for each patient. All participants in our trial were randomly divided into the NB group and the CON group, and we used a sealed envelope for concealed allocation. The ropivacaine and saline, which look identical, were prepared by a nurse who was blinded to the grouping. Another experienced anesthesiologist injected the identical-looking fluid into the SAP and TTMP. Postoperative data collection was recorded by another researcher who was also blinded to group allocation. The enrolled participants and all researchers were unaware of the patients' grouping.

Anesthesia, Surgery, and Analgesia Techniques

Before anesthesia, all patients established appropriate venous access and were monitored with five-lead electrocardiography, noninvasive blood pressure, and peripheral oxygen saturation. The induction of general anesthesia was achieved with 0.03 mg/kg of midazolam, 0.3 µg/kg of sufentanil, 2–3 mg/kg of propofol, and 0.4 mg/kg of rocuronium. After tracheal intubation, we used remifentanyl combined with propofol for the maintenance of anesthesia and the bispectral index (BIS) was maintained between 45 and 55 during surgery. We adjusted the remifentanyl infusion according to the BIS, heart rate, and blood pressure of patients. The children in the NS group received ultrasound-guided TTMP block and SAPB with ropivacaine, and patients in the CON group got the same block with saline after the induction of general anesthesia. After the surgery, all patients were sent to the post-anesthesia care unit (PACU) as scheduled.

All Nuss procedures were performed by the same group of experienced surgeons in our study. Patient-controlled analgesia with intravenous sufentanil but no other opioids was used to perform postoperative analgesia, and patients received acetaminophen 16 mg/kg 3 times a day at most as a supplementary analgesic according to the demands of the children at 48 h after surgery. After discharge, all patients were taking codeine tablets according to their own needs.

TTMP Block and SAPB Techniques

We performed TTMP block and SAPB with the high-frequency linear ultrasound probe (Mindray, Shenzhen, China) with the patients in the supine position after endotracheal intubation. After appropriate sterile preparation, the ultrasound probe was situated parallel to the T4–T5 rib interspace. Then the pectoralis major internal muscle, intercostal muscle, and transversus thoracis muscle were found. The needle (Tuoren, Henan, China) was inserted through the pectoralis major internal muscle and intercostal

muscle and advanced to the interfascial plane between the intercostal muscle and transversus thoracis muscle [18] with 10 ml of 0.2% ropivacaine. SAPB was performed through the midclavicular line at the level of the T4–5 intercostal space, and the same needle was advanced into the interfascial plane between the serratus anterior and latissimus dorsi muscles [19] with an injection of 10–20 mL of ropivacaine (for a patient height < 150 cm, 10 mL of ropivacaine were given; for a patient height between 150 and 160 cm, 15 mL of ropivacaine was given; for patient height > 160 cm, 20 mL of ropivacaine were given). The same nerve block methods were repeated on the other side. The total dose of ropivacaine in all participants did not exceed 3 mg/kg.

Outcome Measurements

The primary endpoint of our study was postoperative pain at 2, 4, 8, 16, 24, 36, and 48 h, which was measured using the Numerical Rating Scale (NRS) score, from 0 (no pain) to 10 (worst severe pain), at rest and movement (during coughing). The secondary outcomes included the intraoperative dosage of remifentanyl, the time to extubation, and the length of stay in the PACU, the total acetaminophen and codeine tablet consumption, time to first bowel movement, time to first flatus, and opioid-related adverse events and the length of hospital stay.

Sample Size Calculation and Statistical Analysis

The sample size of the study was calculated on the basis of our preliminary research ($n=6$ in each group), and we compared postoperative pain, which was measured using the Numerical Rating Scale (NRS) score, from 0 (no pain) to 10 (worst severe pain), at 8 h in two groups (1.1 ± 0.3 in the CN group versus 2.9 ± 0.7 in the CON group).

According to these data, an estimated sample size of 60 patients were needed with a type I error of $\alpha=0.05$, a type II error of $\beta=0.1$, and a power of 90%. Considering a dropout rate of

20%, we included 20% additional patients for the final sample size ($n=36$ in each group).

Statistical analysis was performed using SAS statistical software (version 9.1.3, North Carolina, USA). Continuous data with the two groups were statistically analyzed using the Student *t* test/Mann–Whitney *U* test. The qualitative data were analyzed using Chi-square or Fisher's exact test. Pain intensity at rest and movement was compared with repeated-measures (two-way) analysis of variance. *P*-values less than 0.05 were considered statistically significant.

RESULTS

A total of 72 patients were enrolled in our study. Five cases were excluded from the trial due to refused follow-up (2) and missing data (3). Finally, 67 patients were analyzed in our study, with 33 in the NB group and 34 in the CON group (Fig. 1). No complications associated with the TTMP blocks and SAPB occurred in our study. The groups had no statistically significant difference regarding the perioperative baseline characteristics (Table 1).

Patients in the NB group had significantly lower NRS pain scores at 2, 4, 8, 16, and 24 h after surgery at rest and movement (during coughing) compared with the CON group, and there were no differences at 36 and 48 h after the Nuss procedures between the two groups (Figs. 2, 3). The NB group required significantly less postoperative acetaminophen consumption and lower dosages of intraoperative and postoperative sufentanyl compared with the CON group (Table 2). The intraoperative dosage of remifentanyl in the NE group was significantly lower than that in the CON group (Table 2). The length of stay in the PACU and time to extubation were significantly increased in the CON group compared with the NE group (Table 2). Time to first bowel movement and time to first flatus was earlier in the NB group (Table 2). But there were no significant differences between the groups in terms of the length of hospital stay and codeine tablet consumption (Table 2).

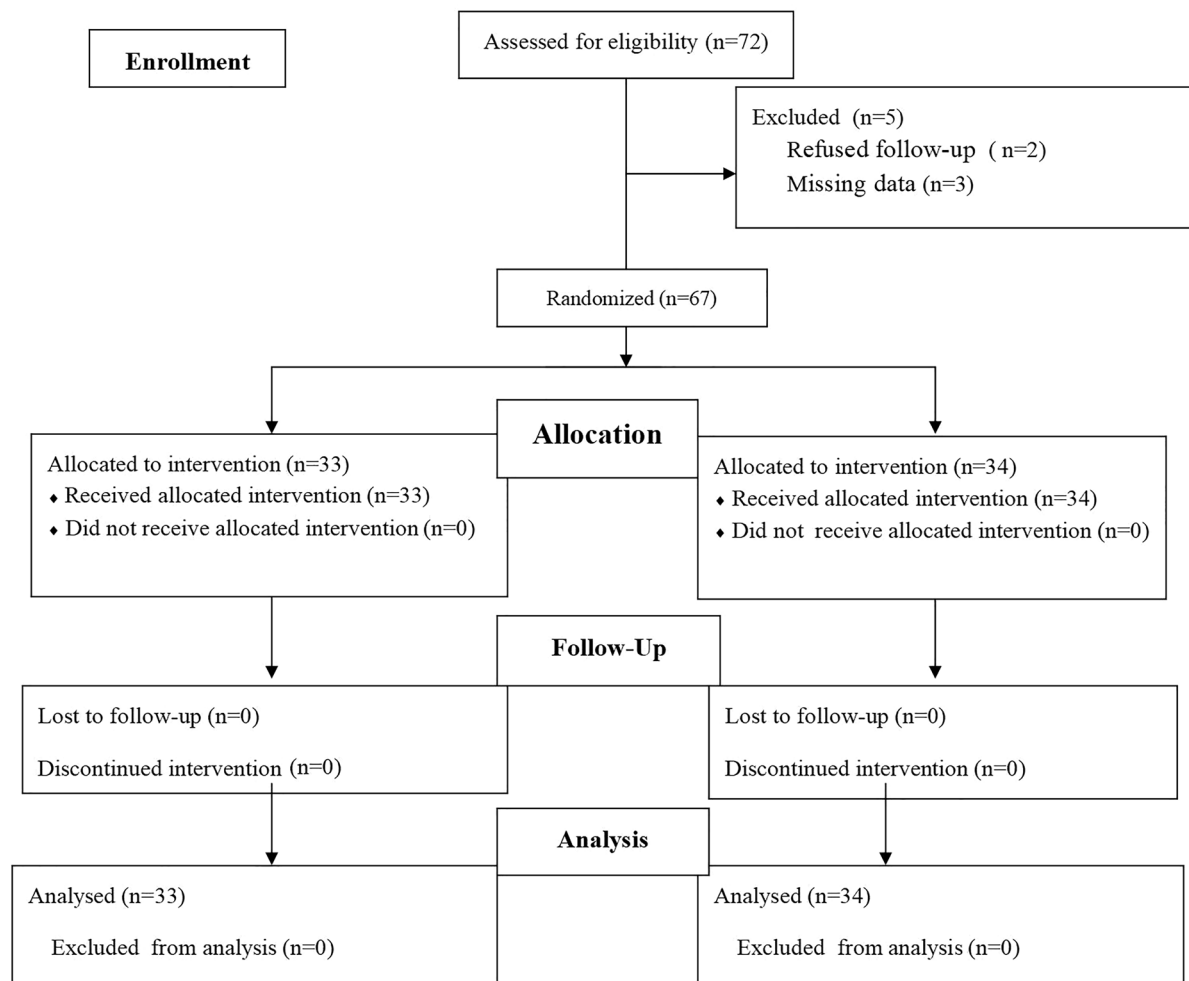


Fig. 1 Patient flow diagram

DISCUSSION

The present study demonstrated that ultrasound-guided TTMP block combined with SAPB for patients undergoing the Nuss procedure provided effective analgesic efficacy for 24 h. What is more, it could also reduce perioperative opioid consumption, time to extubation, and length of stay in the PACU and promote postoperative gastrointestinal function recovery for patients.

Thoracic epidural anesthesia was the gold standard for postoperative pain management in the Nuss procedure before [20]. However, the risk of potential permanent neurological injury, epidural hematoma, abscess, and

hemodynamic instability have restricted its application in the Nuss procedure [21]. Ultrasound-guided bilateral thoracic paravertebral block [22] and the erector spinae plane block [12] also provided effective postoperative analgesia for pectus excavatum patients and reduced the incidence of postoperative behavioral disturbance [11]. Placement in a lateral decubitus position to perform thoracic paravertebral block and the erector spinae plane block was difficult and risky (e.g., falling off the operating table, hemodynamic instability, or tracheal tube displacement or detachment). And nerve injury (spinal nerve and spinal cord), pneumothorax, pleural puncture, or

Table 1 Demographic data and surgical procedures

	NB group (<i>n</i> = 33)	CON group (<i>n</i> = 34)	<i>P</i> -value
Age (years)	13.1 ± 4.6	12.8 ± 4.9	0.53
Weight (kg)	56.9 ± 18.5	55.8 ± 17.9	0.75
Height (cm)	161.2 ± 19.2	158.9 ± 17.1	0.62
BMI (kg/m ²)	21.9 ± 4.9	22.3 ± 5.1	0.47
Sex (male/female)	20/13	19/15	0.59
Associated nerve block time (min)	16.4 ± 5.1	15.9 ± 4.8	0.47
Duration of surgery (min)	80.6 ± 11.8	78.5 ± 10.9	0.67
Number of bar(s) (1/2)	33(100%)/0	34(100%)/0	0.68
ASA			0.71
I	28(84.8%)	27(79.4%)	
II	5(15.2%)	7(20.6%)	

ASA, American Society of Anesthesiologists Physical Status Classification level; BMI, body mass index

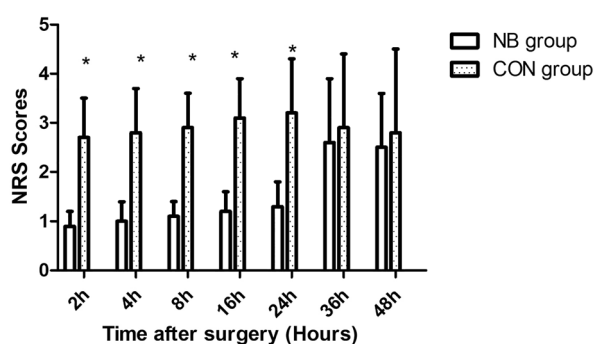


Fig. 2 Pain intensity at rest after surgery, which was measured by NRS. **P* < 0.05 considered statistically significant

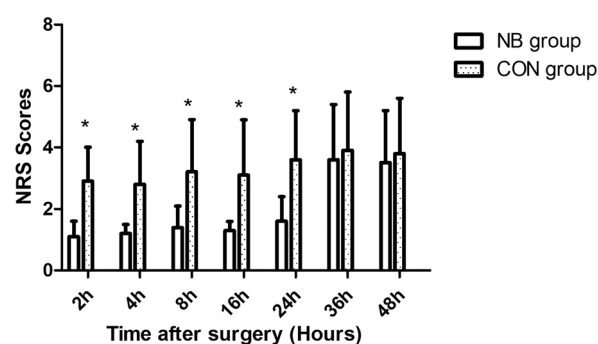


Fig. 3 Pain intensity at movement after surgery, which was measured by NRS. **P* < 0.05 considered statistically significant

vascular puncture may occur during the blocking process.

The serratus anterior plane block [13, 23] and pectoral nerve block [14] (PECSII) were associated with decreased opioid consumption and NRS pain scores for the first day after surgery. SAPB and PECSII were superficial and safe for patients undergoing the Nuss procedure, but they cannot cover anterior branches of intercostal nerves to relieve pain at the site of the internal mammary area. So, we should add a safe nerve block to cover the internal mammary area.

The transversus thoracic muscle plane (TTMP) block covers the internal mammary area and blocks the anterior branches of intercostal nerves [24]. Therefore, we believe that the combination of SAPB and TTMP block may provide effective analgesia for the Nuss procedure. To the authors' knowledge, this was the first randomized, controlled study of which the authors are aware to evaluate SAPB and TTMP block in the Nuss procedure.

Patients undergoing the Nuss procedure usually have severe postoperative pain due to the

Table 2 Intra- and postoperative clinical outcomes

	NB group (<i>n</i> = 33)	CON group (<i>n</i> = 34)	<i>P</i> -value
Intraoperative sufentanyl consumption (μg)	15.6 $\pm$ 3.2	25.3 $\pm$ 5.4	< 0.05
Intraoperative remifentanyl consumption (μg)	764.5 $\pm$ 56.2	1050.7 $\pm$ 156.9	< 0.05
Postoperative sufentanyl consumption (μg)	57.6 $\pm$ 17.5	107.7 $\pm$ 25.6	< 0.01
Acetaminophen consumption (mg/kg)	16.0 $\pm$ 16.0	48.0 $\pm$ 16.0	< 0.01
Time to extubation (min)	11.5 $\pm$ 6.7	25.6 $\pm$ 11.9	< 0.05
Length of stay in the PACU (min)	30.2 $\pm$ 10.5	60.3 $\pm$ 18.3	< 0.05
Length of hospital stay (h)	96.3 $\pm$ 15.7	100.4 $\pm$ 20.2	0.78
Codeine tablet consumption (mg)	7.8 $\pm$ 3.2	8.4 $\pm$ 2.9	0.86
Time to first flatus (h)	15.8 $\pm$ 4.2	21.8 $\pm$ 5.9	< 0.05
Time to first bowel movement (h)	20.3 $\pm$ 7.9	31.7 $\pm$ 9.9	< 0.05

force from the Nuss bar elevating the sternal depression and correcting the deformity of the chest wall, incision in the rich innervation of the chest wall, and muscular spasms [25, 26].

The SAPB could block the sensory dermatomes of T2–T9 by affecting the thoracic intercostal nerves [27], and the TTMP block could block the sensory dermatomes of T2–T6 by affecting the anterior branches of the intercostal nerves [28]. So the SAPB combined with TTMP blocks completely covered the forced area and the incision area.

Due to the effective analgesia of combined nerve blocks, patients in the NB group need less sufentanyl and remifentanyl during surgery. This might be the reason for the earlier extubation and shorter length of PACU stay. Patients in the NB group had lower NRS pain scores both at rest and after movement, and less sufentanyl and acetaminophen consumption after the Nuss procedure. The decrease in perioperative opioid dosage may be the main reason for the rapid recovery of gastrointestinal function in the NB group.

The potential complications of SAPB and TTMP blocks included infection, pneumothorax, ropivacaine allergy, hematoma, and vascular injury. However, all patients in our study had no adverse side effects, indicating

that the blocks are safe and effective in the Nuss procedure.

The study had some limitations. Continuous nerve blocks or adding ultra-long-acting local anesthetics for nerve block may provide longer effective analgesia. Therefore, it is necessary to explore the use of continuous nerve blocks or adding ultra-long-acting local anesthetics in patients undergoing the Nuss procedure in future. The concentration of ropivacaine in all patients was based on other research. The small sample size limited generalizability, and further research is needed to determine the optimal ropivacaine concentration. The control group patients should have received SAPB, and we will do this in our future study.

CONCLUSION

The present study showed that bilateral SAPB combined with TTMP block could provide effective analgesia for 24 h after surgery, reduce perioperative sufentanyl and remifentanyl consumption, promote early recovery of intestinal function, and decrease the time to extubation and the length of stay in the PACU in patients undergoing the Nuss procedure.

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Author Contributions. Yang Zhang and Shibiao Chen conceived and designed this study and collected the data. Tao Chen and Yu Xu were responsible for study execution and manuscript writing. Tao Chen, Yu Chen, and Shibiao Chen were responsible for data analysis. All authors have read and approved the final version of the manuscript.

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Data Availability. The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest. Tao Chen, Yu Xu, Yu Chen, Yang Zhang, and Shibiao Chen have no conflict of interest.

Ethical approval. This single-center, observational, prospective clinical study was approved by the ethics committee of First Affiliated Hospital of Nanchang University, and written informed consent was obtained from all subjects participating in the trial. Our study was registered in the Chinese Clinical Trial Registry (ChiCTR2000038506). Our study was performed in accordance with the Helsinki Declaration of 1964 and its later amendments.

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